Scalable and Interpretable Bayesian Methods for Spatial Data Analysis

Autoencoder-Based Data Fusion and Minibatch GP Fitting

Jacob Johnson

Spatial Statistics & GP Reading Group
February 12, 2025



Background



Alternate Title: The work that got me kicked out of a Master's program.

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BYU Statistics

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Fusing Climate Data Products Using a Spatially Varying Autoencoder

- Jacob A. Johnson, Matthew J. Heaton, William F. Christensen, Lynsie R. Warr, and Summer B. Rupper

Minibatch Markov Chain Monte Carlo Algorithms for Fitting Gaussian Processes

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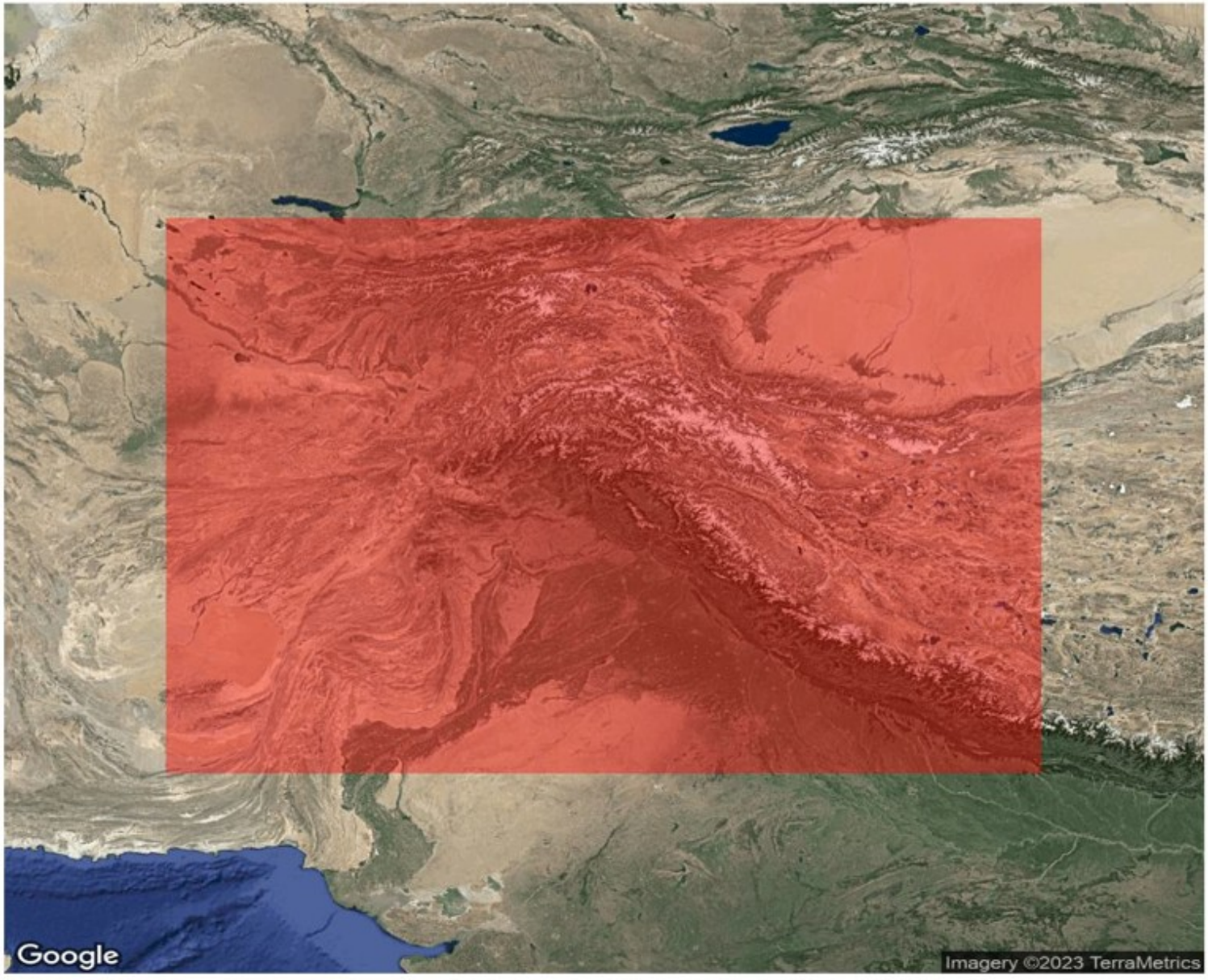
Problem Background



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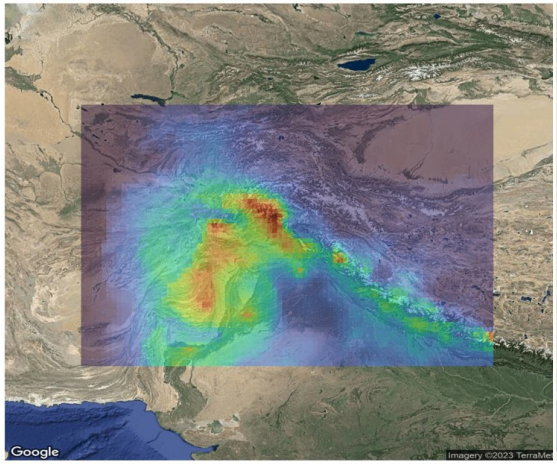
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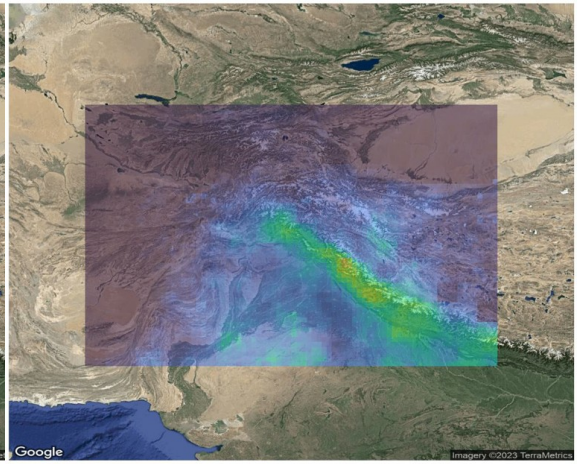


Problem Background

APHRODITE

Asian Precipitation Highly-Resolved Observational Data Integration Towards Evaluation of Water Resources



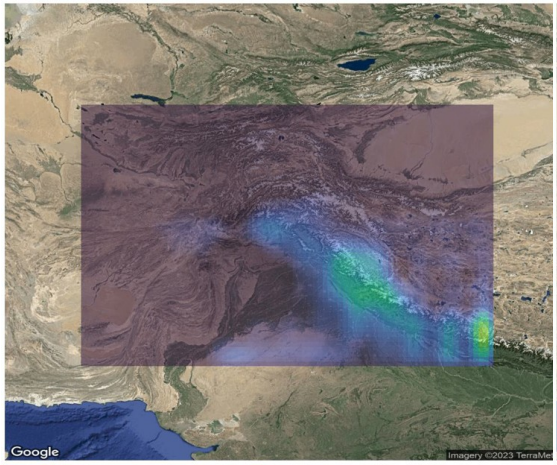


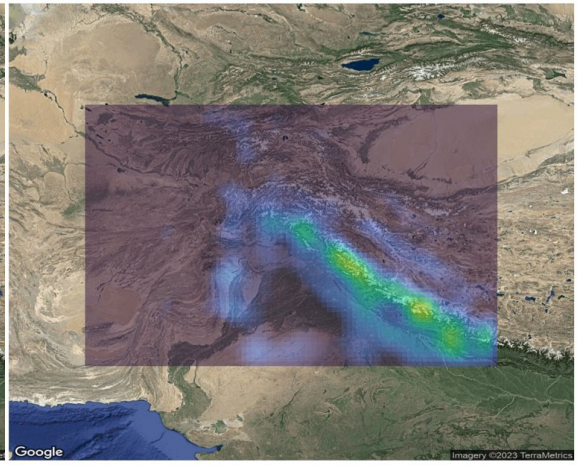
TRMM

Tropical Rainfall Measuring Mission

MERRA2

Modern-Era Retrospective analysis for Research and Applications, Version 2





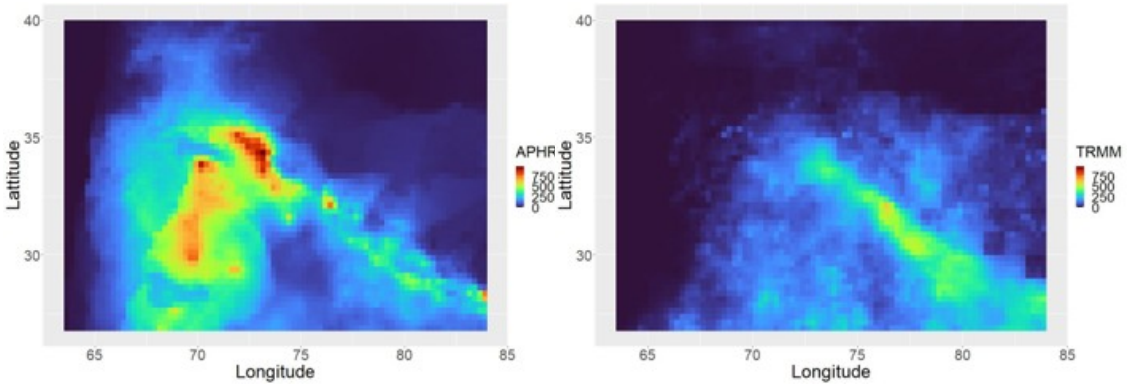
ERA5

5th Generation ECMWF atmospheric reanalysis

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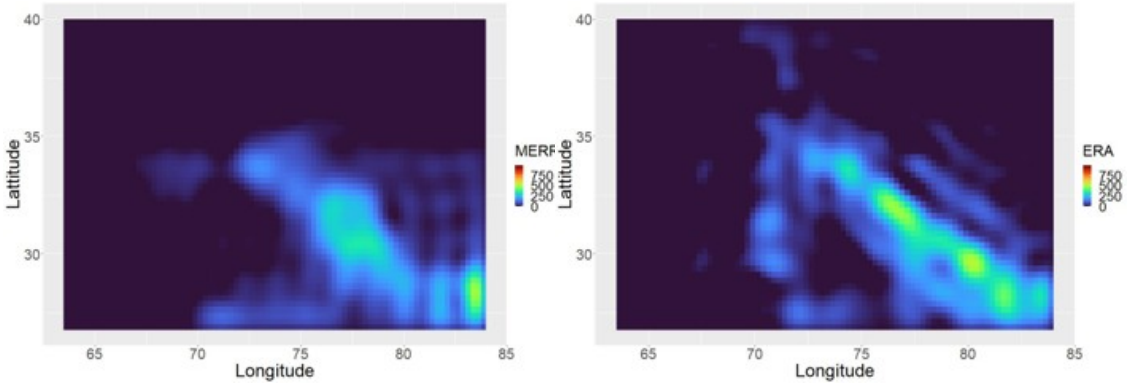


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Current Approaches



- Flat Averaging
  - Consensus created from an equally weighted linear combination of the 4 data products - no estimation required

[1] Christensen et al. [2019]
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- Factor Analysis Approach[1]
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  - Consider each individual data product as a noisy representation of a spatially smooth consensus.
  - Consensus is a linear combination of the individual data products with estimated weights
- Spatial Factor Analysis Approach[2]
  - Consensus is a spatially-varying linear combination of the individual data products
  - Spatially-varying weights are estimated

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[2] Neeley et al. [2014]

Research Goal



Develop a novel way to fuse/meld these differing data products that:

- includes spatial structure,
- considers a non-linear combination,
- maintains interpretability,
- has uncertainty quantification.

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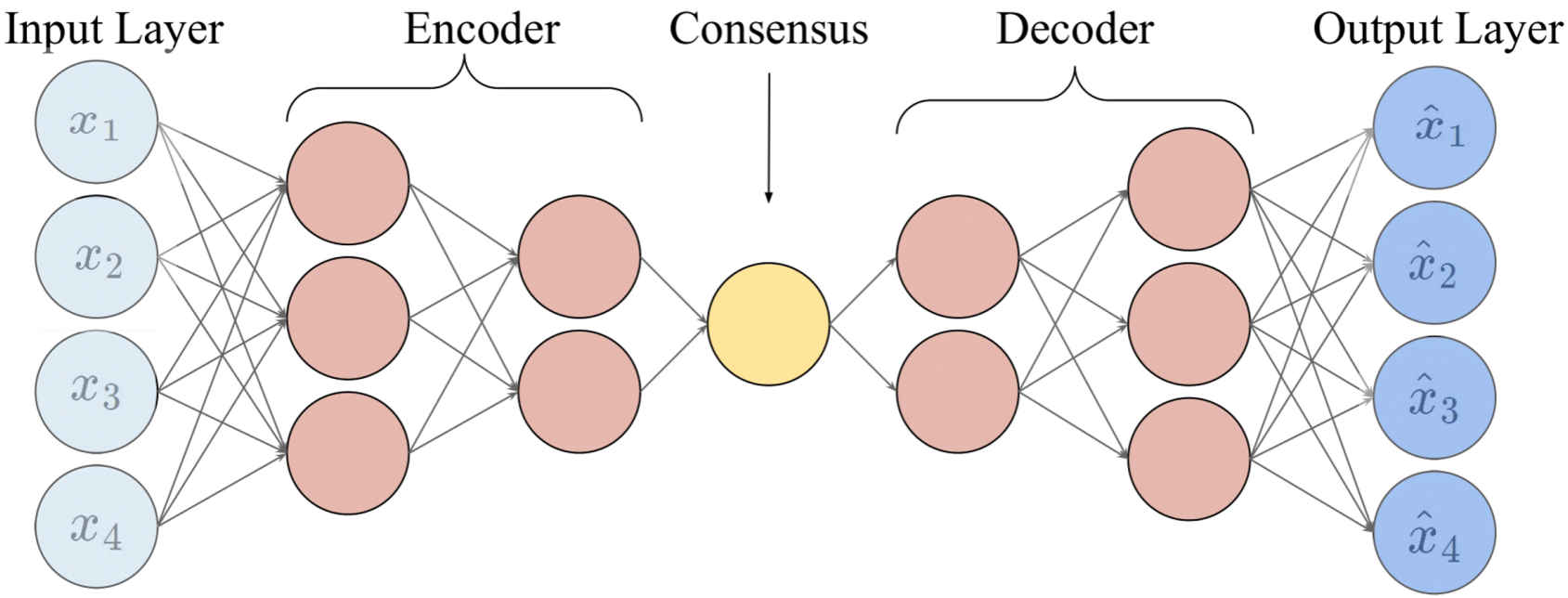


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Our solution: A *Spatially-Varying Bayesian Autoencoder*

Autoencoder Structure



Autoencoder Model



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$$Y_d(\boldsymbol{s}) = g(Z_d(\boldsymbol{s})) \tag{1}$$

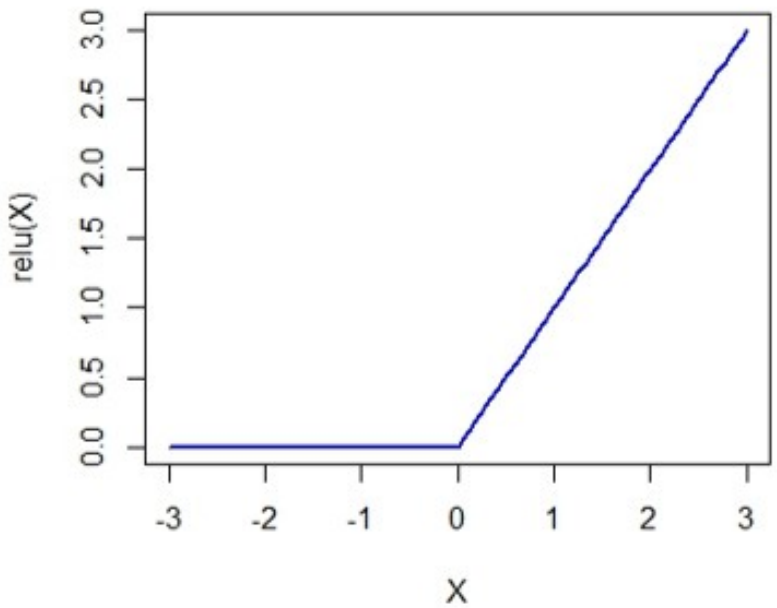
where $g(\cdot)$ is some transformation function that preserves the support of the response and $Z_d(\boldsymbol{s})$ is a latent variable.

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$$Y_d(\boldsymbol{s}) = g(Z_d(\boldsymbol{s})) \tag{1}$$

Next, we assume that

$$Z_d(\boldsymbol{s}) = f_d(\boldsymbol{s}) + \varepsilon, \qquad \varepsilon \overset{iid}{\sim} \mathcal{N}(0, \sigma^2). \tag{2}$$

We will use an autoencoder as our non-linear function f so,

$$Z_d(\boldsymbol{s}) = N_d^{\text{out}}(\boldsymbol{s}) + \varepsilon, \qquad \varepsilon \overset{iid}{\sim} \mathcal{N}(0, \sigma^2). \tag{3}$$

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Next, the structure of the autoencoder with P_ℓ inputs at layer $\ell \in \{1, \ldots, L\}$ is

$$\boldsymbol{N}^\ell(\boldsymbol{s}) = a_\ell\left(\boldsymbol{w}_0^\ell(\boldsymbol{s}) + \boldsymbol{W}^\ell(\boldsymbol{s})\boldsymbol{N}^{\ell-1}(\boldsymbol{s})\right) \tag{3}$$

- $a_\ell(\cdot)$ is an activation function
- $\boldsymbol{w}_0^\ell$ is a P_ℓ vector of biases
- $\boldsymbol{W}^\ell(\boldsymbol{s})$ is a $P_\ell \times P_{\ell-1}$ matrix of weights for transitioning from $\ell - 1$ to ℓ

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$$\boldsymbol{N}^{\ell}(\boldsymbol{s}) = a_{\ell}\left(\boldsymbol{w}_0^{\ell}(\boldsymbol{s}) + \boldsymbol{W}^{\ell}(\boldsymbol{s})\boldsymbol{N}^{\ell-1}(\boldsymbol{s})\right) \tag{3}$$

For completeness, the output and input layers are given by

$$\boldsymbol{N}^{\text{out}}(\boldsymbol{s}) = a\left(\boldsymbol{w}_0^{\text{out}}(\boldsymbol{s}) + \boldsymbol{W}^{\text{out}}(\boldsymbol{s})\boldsymbol{N}^{L}(\boldsymbol{s})\right) \tag{4}$$

$$\boldsymbol{N}^{0}(\boldsymbol{s}) = (Y_1(\boldsymbol{s}), \ldots, Y_d(\boldsymbol{s}))' \tag{5}$$

Interpetability Constraints



In order to obtain an interpretable consensus, we introduce some additional constraints. First, we set

$$\boldsymbol{w}_0^\ell \equiv \mathbf{0} \text{ for all } l \text{ (including the output)} . \tag{6}$$

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Second, we assume a lower triangular structure for each weight matrix in the decoder portion of the autoencoder. For example,

$$\boldsymbol{W}^{\text{out}}(s) = \begin{pmatrix} 1 & 0 & 0 \\ \left(\boldsymbol{W}^{\text{out}}(\boldsymbol{s})\right)_{2,1} & 1 & 0 \\ \left(\boldsymbol{W}^{\text{out}}(\boldsymbol{s})\right)_{3,1} & \left(\boldsymbol{W}^{\text{out}}(\boldsymbol{s})\right)_{2,1} & 1 \\ \left(\boldsymbol{W}^{\text{out}}(\boldsymbol{s})\right)_{4,1} & \left(\boldsymbol{W}^{\text{out}}(\boldsymbol{s})\right)_{4,2} & \left(\boldsymbol{W}^{\text{out}}(\boldsymbol{s})\right)_{4,3} \end{pmatrix} \tag{7}$$

This constraint *ties down* the scale of the consensus to match one of the data products without constraining the autoencoder to reproduce the output directly[3].

[3] Neeley et al. [2014]

Spatially Smooth Weights



We enforce spatial structure by representing the weights using spatial basis functions (e.g. kernels, predictive processes or bisquare functions [4]).

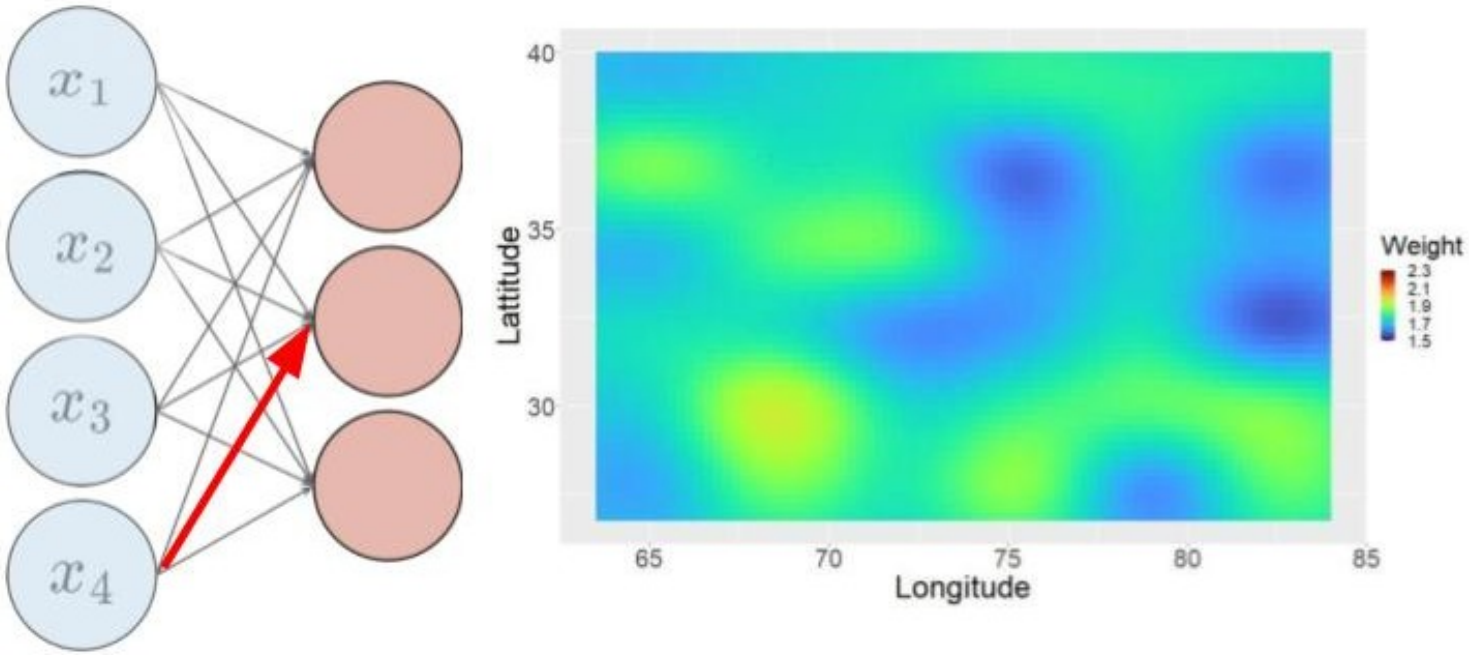
$$\left(\boldsymbol{W}^{\ell}(\boldsymbol{s})\right)_{p,j} = \boldsymbol{b}'(\boldsymbol{s})\boldsymbol{\theta}^{l}_{p,j}, \qquad \boldsymbol{b}(\boldsymbol{s}) = (1, b_1(\boldsymbol{s}), \ldots, b_K(\boldsymbol{s}))' \tag{8}$$

[4] see Cressie et al. [2022]

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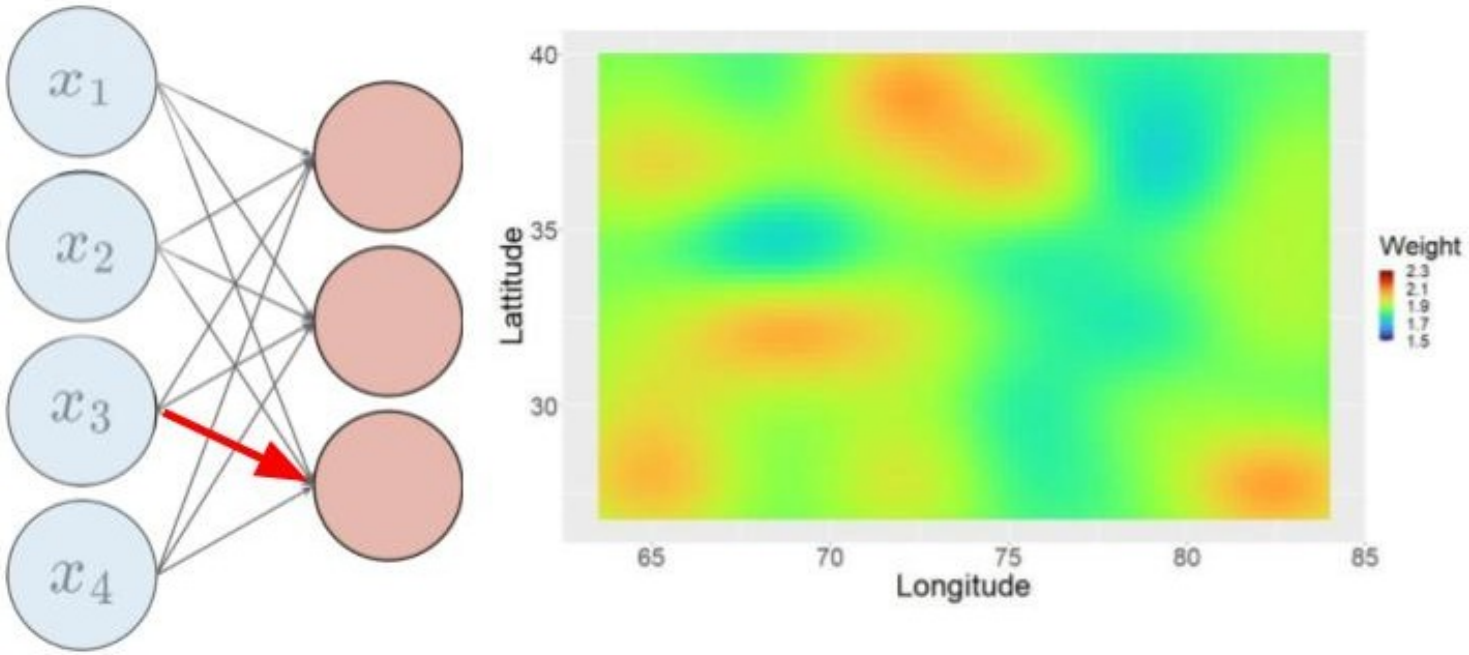


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Priors and Model Fitting



We place diffuse priors on the basis coefficients and σ^2.

$$\left(\boldsymbol{\theta}^l_{p,j}\right)_k \overset{iid}{\sim} \mathcal{N}(0, 5)$$

$$\sigma^2 \sim \text{Inverse Gamma}(2.1, 1.1)$$

Given the above priors, inference is carried out via Markov chain Monte Carlo (MCMC) sampling from the posterior distribution, using adaptive Metropolis steps [5] to update the basis coefficients.

[5] Haario et al. [2001]

Full Bayesian Autoencoder Model

$$Y_d(\boldsymbol{s}) = g(Z_d(\boldsymbol{s}))$$

$$Z_d(\boldsymbol{s}) = N_d^{\text{out}}(\boldsymbol{s}) + \varepsilon, \quad \varepsilon \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$$

$$\boldsymbol{N}^{\ell}(\boldsymbol{s}) = a_{\ell}\left(\boldsymbol{w}_0^{\ell}(\boldsymbol{s}) + \boldsymbol{W}^{\ell}(\boldsymbol{s})\boldsymbol{N}^{\ell-1}(\boldsymbol{s})\right)$$

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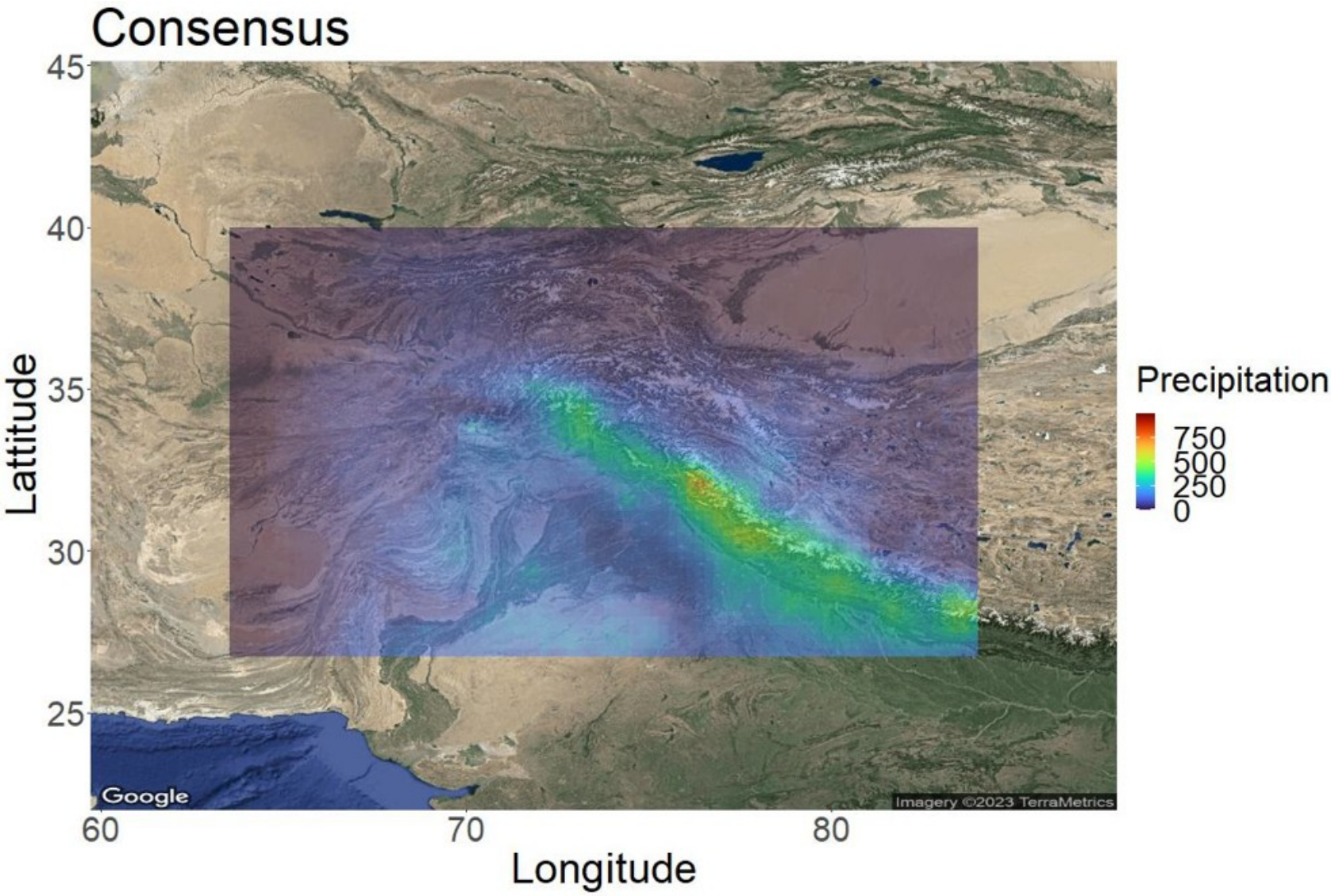
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$$\left(\boldsymbol{W}^{\ell}(\boldsymbol{s})\right)_{p,j} = \boldsymbol{b}'(\boldsymbol{s})\boldsymbol{\theta}_{p,j}^{l}$$

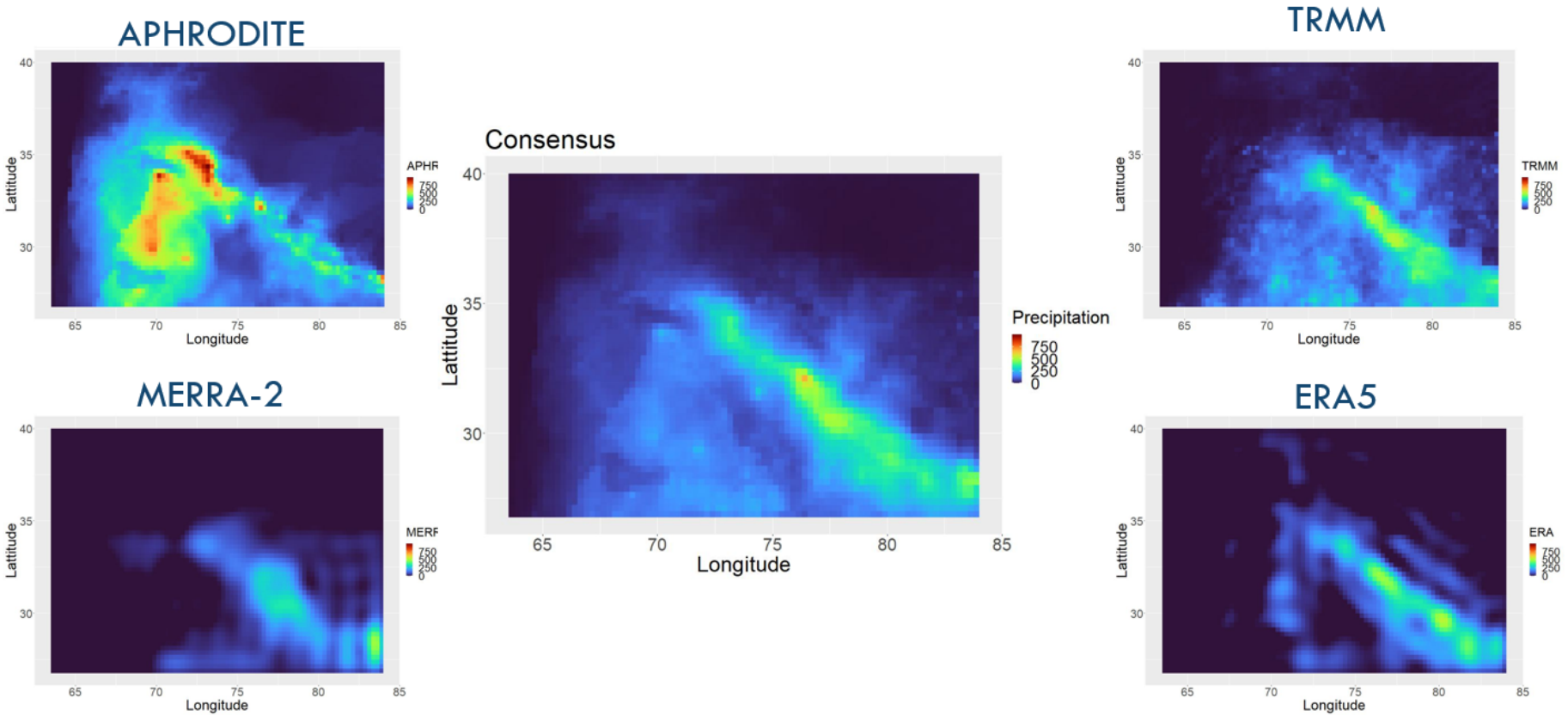
$$\left(\boldsymbol{\theta}_{p,j}^{l}\right)_k \overset{iid}{\sim} \mathcal{N}(0, 5)$$

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Consensus Results

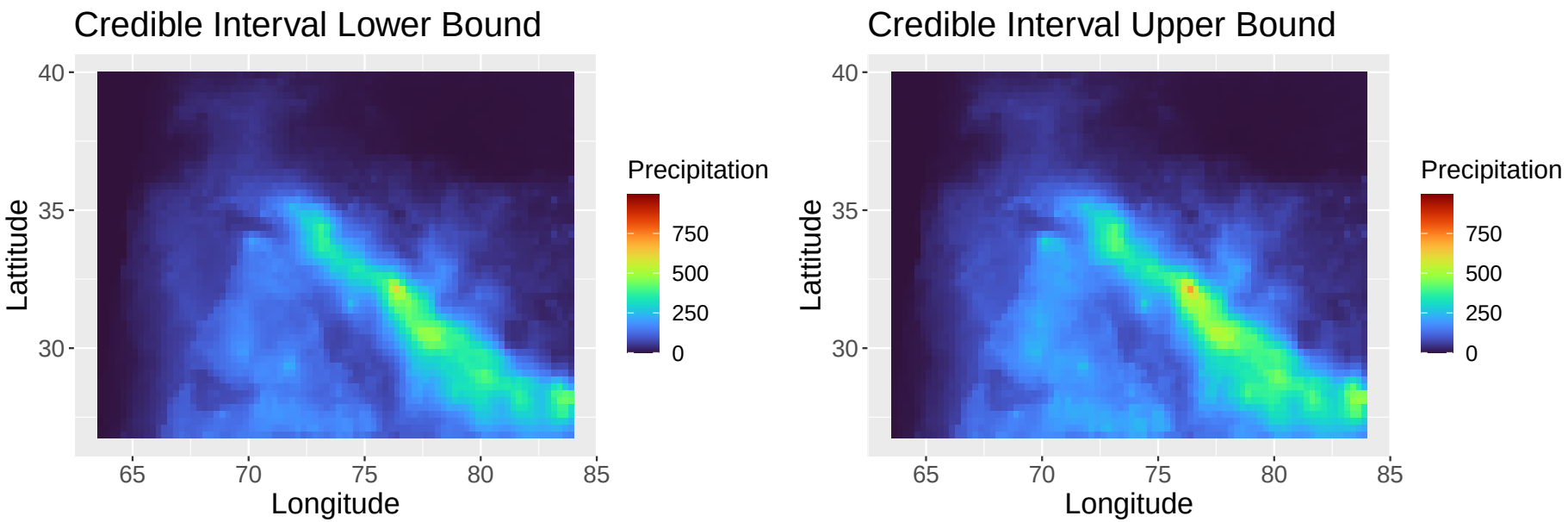


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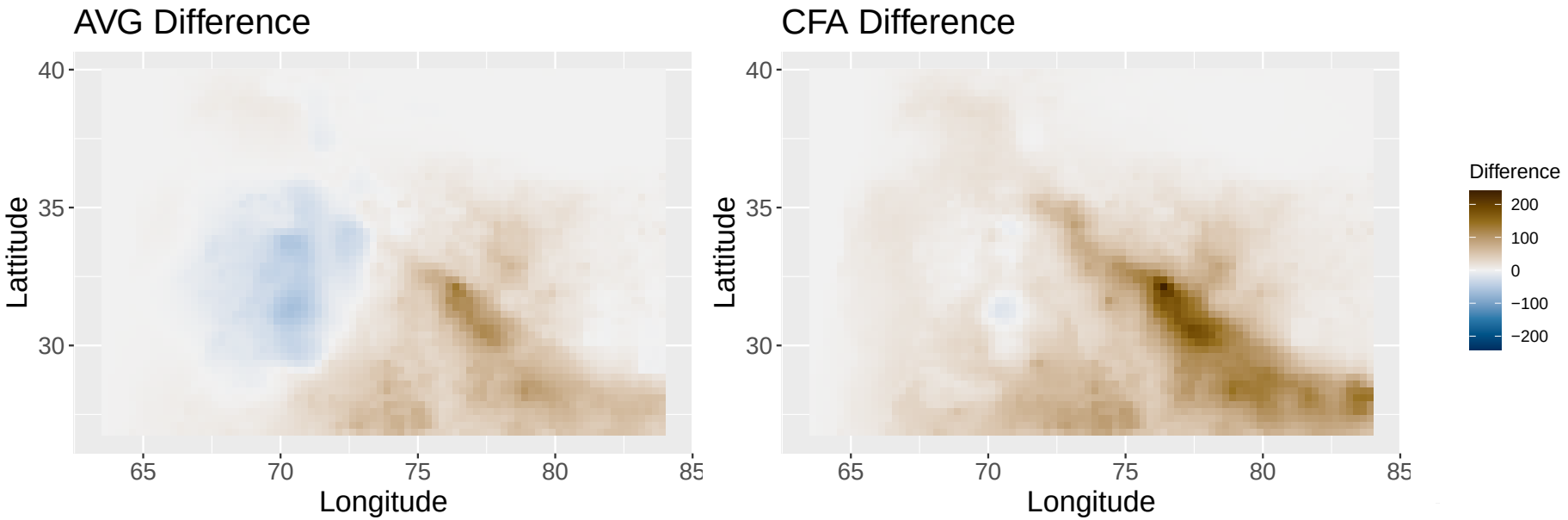
Consensus Uncertianty Quantification

A major advantage to using Bayesian methods is the inclusion of uncertainty quantification about model parameters and predictions. Using the saved MCMC draws, we can easily construct credible intervals and calculate probabilities of interest on the merged consensus.



Comparison to Other Approaches

In comparing our autoencoded product to other approaches, it is important to note that because there is no definitive "ground truth" which we cancompare to, we cannot make a claim that our autoencoded product is "better" or "worse" than another fused product.



We can however point out the differences between our consensus and the results from other methods. Regions where our consensus is higher are shown in brown, while areas where it was lower are shown in blue.

Measuring Contribution

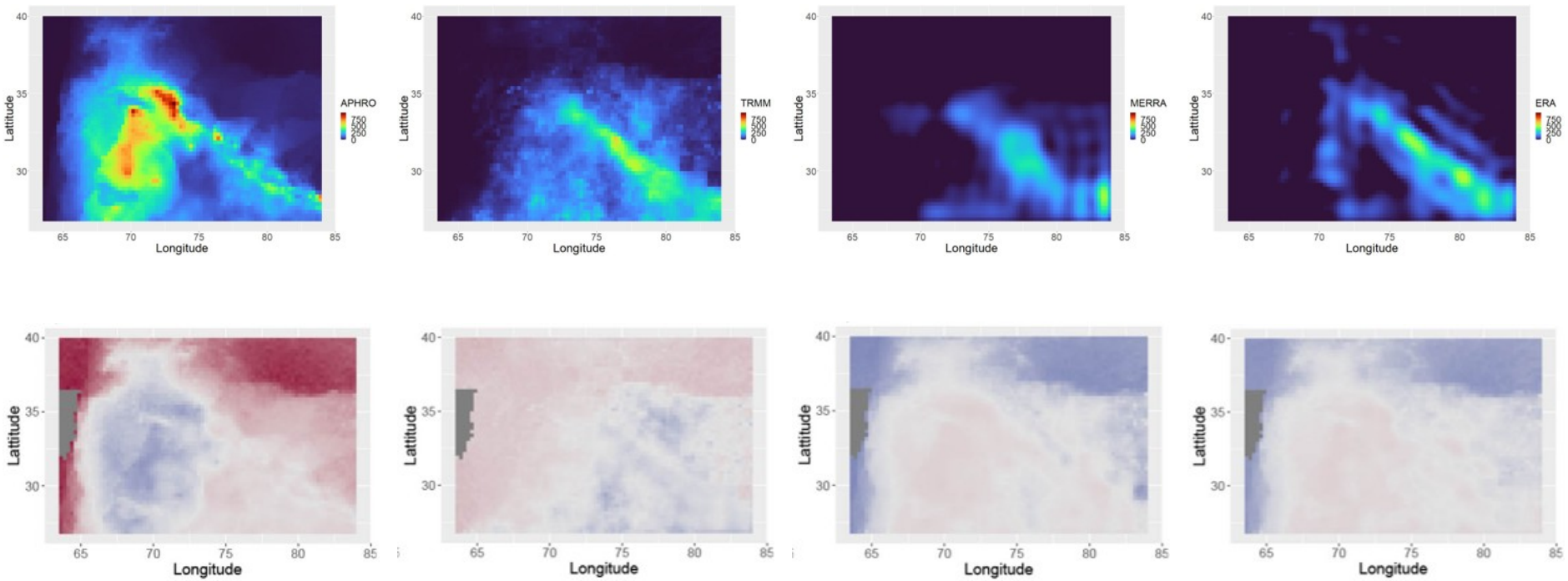


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APHRODITE **TRMM** **MERRA-2** **ERA5**



Summary



In this research, we developed a novel approach to data fusion using a spatially-varying autoencoder.

- By constraining the autoencoder weights, we created a consensus data product that can be interpreted as a valid data product.
- The fused consensus from our autoencoder captures different spatial features than those found using previous approaches to data fusion
- By using the Bayesian paradigm, we were able to include uncertainty quantifications on all model parameters and on the fused consensus.

[6] see https://github.com/jajcelloplayer/Weather-Autoencoder

Summary



In this research, we developed a novel approach to data fusion using a spatially-varying autoencoder.

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- By using the Bayesian paradigm, we were able to include uncertainty quantifications on all model parameters and on the fused consensus.

The HMA data we used include nearly 20 years of monthly data. While we successively applied the autoencoder to each month independently [6], accounting for temporal correlation between months may also be of interest.

[6] see https://github.com/jajcelloplayer/Weather-Autoencoder

Minibatch Markov Chain Monte Carlo Algorithms for Fitting Gaussian Processes

Matthew J. Heaton and Jacob A. Johnson

Problem Background



Gaussian processes (GPs) are a highly flexible, nonparametric statistical model commonly used to account for correlation between observations. However, the computational load of fitting a Gaussian process is $\mathcal{O}(n^3)$ making them infeasible for large datasets.

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We outline both approximate and exact minibatch MCMC algorithms that substantially reduce the computation of fitting a GP by only considering small subsets of the data at a time.

Preliminaries: Gaussian Processes



Let $Y(\boldsymbol{s})$ be a response variable measured at location $\boldsymbol{s} \in \mathcal{D} \subset \mathbb{R}^d$.
$Y(\boldsymbol{s})$ is said to follow a Gaussian process (GP) if for any finite collection of locations $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_n$

$$\boldsymbol{Y} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \tag{9}$$

where

- $\boldsymbol{Y} = (Y(\boldsymbol{s}_1), \ldots, Y(\boldsymbol{s}_n))'$
- $\boldsymbol{\mu} = (\mu(\boldsymbol{s}_1), \ldots, \mu(\boldsymbol{s}_n))'$ (Mean vector)
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We consider the specific case where

$$\boldsymbol{\mu} = \boldsymbol{X}\boldsymbol{\beta} \text{ and } \boldsymbol{\Sigma} = \sigma^2 \boldsymbol{R} = \sigma^2(\omega \boldsymbol{I} + (1-\omega)\boldsymbol{M}) \tag{10}$$

so that the covariance is governed by parameters $\boldsymbol{\phi}$ such that $\boldsymbol{M}_{ij} = \rho(\boldsymbol{s}_i, \boldsymbol{s}_j \mid \boldsymbol{\phi})$.

The Conditional Likelihood



The likelihood for $\boldsymbol{\Theta} = (\boldsymbol{\beta}, \sigma^2, \omega, \boldsymbol{\phi}')'$ can be written as a series of conditionals:

$$\mathcal{L}(\boldsymbol{\Theta}) = f_1(Y(\boldsymbol{s}_1) \mid \boldsymbol{\Theta}) \prod_{i=2}^{n} f_i(Y(\boldsymbol{s}_i) \mid Y(\boldsymbol{s}_1), \ldots, Y(\boldsymbol{s}_{i-1}), \boldsymbol{\Theta}) \tag{11}$$

where each $f_i(\cdot)$ is a Gaussian PDF with mean μ_i and variance $\sigma^2 v_i$.

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where each $f_i(\cdot)$ is a Gaussian PDF with mean μ_i and variance $\sigma^2 v_i$.

$$\mu_i = \begin{cases} \boldsymbol{x}'(\boldsymbol{s}_1)\boldsymbol{\beta} & \text{if } i = 1 \\ \boldsymbol{x}'(\boldsymbol{s}_i)\boldsymbol{\beta} + \boldsymbol{R}(i, \mathcal{P}_i)\boldsymbol{R}^{-1}(\mathcal{P}_i, \mathcal{P}_i)(\boldsymbol{Y}_{\mathcal{P}_i} - \boldsymbol{X}_{\mathcal{P}_i}\boldsymbol{\beta}) & \text{if } i > 1 \end{cases} \tag{12}$$

$$v_i = \begin{cases} 1 & \text{if } i = 1 \\ 1 - \boldsymbol{R}(i, \mathcal{P}_i)\boldsymbol{R}^{-1}(\mathcal{P}_i, \mathcal{P}_i)\boldsymbol{R}(\mathcal{P}_i, i) & \text{if } i > 1 \end{cases} \tag{13}$$

where $\mathcal{P}_i = \{1, \ldots, i-1\}$ is the set of points preceding observation i.

Vecchia Approximation

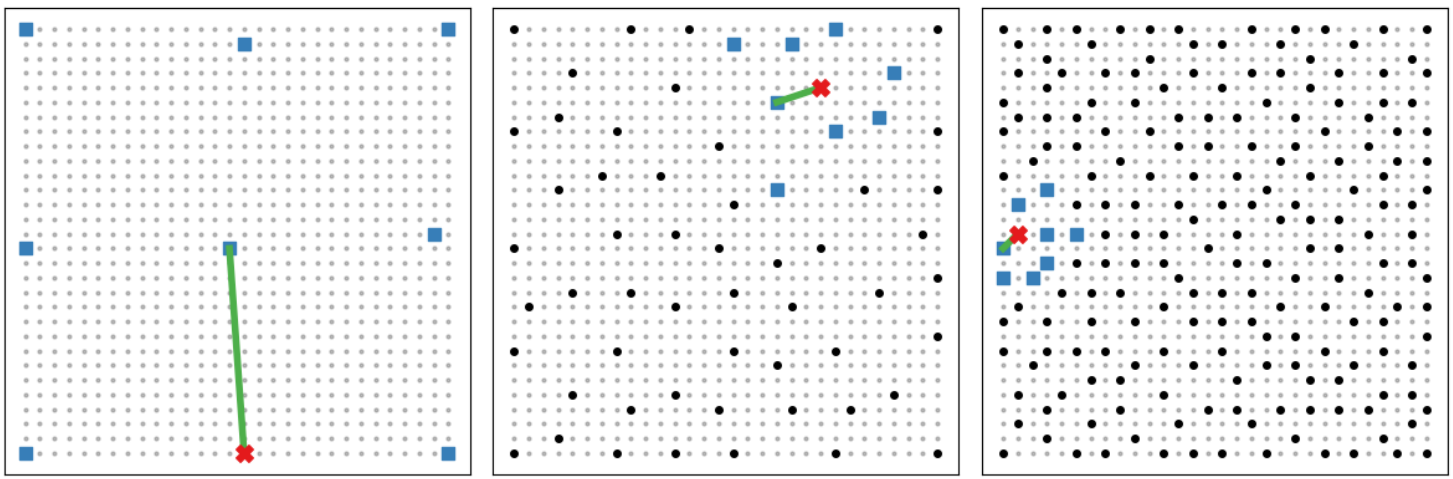


Calculating μ_i and v_i requires inverting large matrices. We adopt the **Vecchia** or **Nearest Neighbor** approximation, replacing the full history $\mathcal{P}_i$ with the set $\mathcal{N}_i$ of the M nearest previously ordered neighbors of $\boldsymbol{s}_i$.

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We construct these $\mathcal{N}_i$ according to *max-min* ordering.



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$$\tilde{\mu}_i = \begin{cases} \boldsymbol{x}'(\boldsymbol{s}_1)\boldsymbol{\beta} & \text{if } i = 1 \\ \boldsymbol{x}'(\boldsymbol{s}_i)\boldsymbol{\beta} + \boldsymbol{R}(i, \mathcal{N}_i)\boldsymbol{R}^{-1}(\mathcal{N}_i, \mathcal{N}_i)(\boldsymbol{Y}_{\mathcal{N}_i} - \boldsymbol{X}_{\mathcal{N}_i}\boldsymbol{\beta}) & \text{if } i > 1 \end{cases} \tag{14}$$

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Conjugate Parameters



We assume priors such that complete conditional distributions of β_p and σ^2 are conjugate with respect to the Vecchia likelihood.

$$\beta_p \mid - \sim \mathcal{N}\left(\left[\frac{\sum_{i=1}^n q_1(\boldsymbol{s}_i)}{\sigma^2} + \frac{1}{s_p^2}\right]^{-1}\left(\frac{\sum_{i=1}^n q_2(\boldsymbol{s}_i)}{\sigma^2} + \frac{m_p}{s_p^2}\right), \ldots\right) \tag{16}$$

$$\sigma^2 \mid - \sim \mathcal{IG}\left(\frac{n}{2} + a_\sigma, \frac{\sum_{i=1}^n q_3(\boldsymbol{s}_i)}{2} + b_\sigma\right) \tag{17}$$

Where q_1, q_2, q_3 are functions of the data and nearest neighbors.

Non-conjugate Parameters

We generally assume $\boldsymbol{\theta} = (\omega, \boldsymbol{\phi}')' \sim \pi(\cdot)$. Due to lack of conjugacy, MCMC sampling of $\boldsymbol{\theta}$ is done via accept-reject algorithms with acceptance rule

$$\Delta(\boldsymbol{\theta}_{\text{prop}}, \boldsymbol{\theta}_{\text{cur}}) = \sum_{i=1}^{n} \Lambda_i - \log\left(\frac{\pi(\boldsymbol{\theta}_{\text{prop}})g(\boldsymbol{\theta}_{\text{cur}} \mid \boldsymbol{\theta}_{\text{prop}})}{\pi(\boldsymbol{\theta}_{\text{cur}})g(\boldsymbol{\theta}_{\text{prop}} \mid \boldsymbol{\theta}_{\text{cur}})}\right) + L \tag{18}$$

where a proposed $\boldsymbol{\theta}_{\text{prop}}$ is accepted if and only if $\Delta(\boldsymbol{\theta}_{\text{prop}}, \boldsymbol{\theta}_{\text{cur}}) > 0$.

- $\Lambda_i = \log[f_i(Y(\boldsymbol{s}_i) \mid \boldsymbol{Y}_{\mathcal{N}_i}, \boldsymbol{\beta}, \sigma^2, \boldsymbol{\theta}_{\text{prop}})/f_i(Y(\boldsymbol{s}_i) \mid \boldsymbol{Y}_{\mathcal{N}_i}, \boldsymbol{\beta}, \sigma^2, \boldsymbol{\theta}_{\text{cur}})]$ is the log-likelihood ratio for observation i
- $g(\cdot)$ is the proposal distribution
- L is a random variable

Minibatch Approximations



We can approximate the sums in the conjugate updates and non-conjugate acceptance rule using a minibatch $\mathcal{B}$ of size B:

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Because our approximation is an average, we assume

$$n\bar{q}_{j,B} \dot{\sim} \mathcal{N}\left(n\bar{q}_{j,n}, \frac{n^2}{B}\sqrt{\frac{n-B}{n-1}}\sigma^2_{q_j}\right), \quad n\bar{\Lambda}_B \dot{\sim} \mathcal{N}\left(n\bar{\Lambda}_n, \frac{n^2}{B}\sqrt{\frac{n-B}{n-1}}\sigma^2_{\Lambda}\right) \quad (21)$$

where $\sigma^2_{q_j} = \mathbb{V}\text{ar}(q_j(\boldsymbol{s}_i))$ and $\sigma^2_{\Lambda} = \mathbb{V}\text{ar}(\Lambda_i)$.

Minibatch Algorithm Approaches



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Minibatch Algorithm Approaches



By replacing the full sums with minibatch approximations, we can do inference and prediction using GPs at a much lower computational cost.
Possible approaches include:

1. Simply replace $\bar{\Lambda}_n$, $\bar{q}_{j,n}$ with $\bar{\Lambda}_B$, $\bar{q}_{j,B}$ and set $L = -\log(U)$ in a minibatch approximation of the Metropolis-Hastings algorithm for a fixed batch size.
2. Account for the fact that $\bar{\Lambda}_n \neq \bar{\Lambda}_B$, and let the algorithm select a batch size based on how much variability will be added by minibatching.

Algorithm 1



Initialize $\boldsymbol{\beta}_{\text{cur}}$, σ^2_{cur}, $\boldsymbol{\theta}_{\text{cur}}$
Set batch size B
Randomly divide the n datapoints into H batches of size B
for epoch in $1, \dots E$ **do**
    for h in $1, \dots, H$ **do**
        for p in $0, \dots P$ **do**
            Draw β_p from (16) using minibatch h
        end for
        Draw σ^2 from (17) using minibatch h
        Propose $\boldsymbol{\theta}_{\text{prop}} \sim g()$
        Let $L = -\log(U)$, $U \sim \mathcal{U}(0, 1)$. Using minibatch h calculate

$$\Delta(\boldsymbol{\theta}_{\text{prop}}, \boldsymbol{\theta}_{\text{cur}}) = n\bar{\Lambda}_B + \log\left(\frac{\pi(\boldsymbol{\theta}_{\text{prop}})g(\boldsymbol{\theta} \mid \boldsymbol{\theta}_{\text{prop}})}{\pi(\boldsymbol{\theta}_{\text{cur}})g(\boldsymbol{\theta}_{\text{prop}} \mid \boldsymbol{\theta})}\right) + L$$

        $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta}_{\text{prop}}$ if $\Delta(\boldsymbol{\theta}_{\text{prop}}, \boldsymbol{\theta}) > 0$ using the approximated acceptance rule
        Store $\boldsymbol{\beta}$, σ^2, $\boldsymbol{\theta}$
    end for
end for

The Approximate Barker Rule



Barker [1965] proposes a rule where L follows a standard logistic distribution. Let L follow a standard logistic distribution and be represented as the sum $L = L_1 + L_2$ where $L_1 \sim \mathcal{N}(0, \sigma^2_{L_1})$ and $L_2 \sim h(\cdot)$ where $h(\cdot)$ is a *correction* distribution from the Gaussian distribution to the standard logistic (estimated via LASSO with positivity constraints).

The Approximate Barker Rule



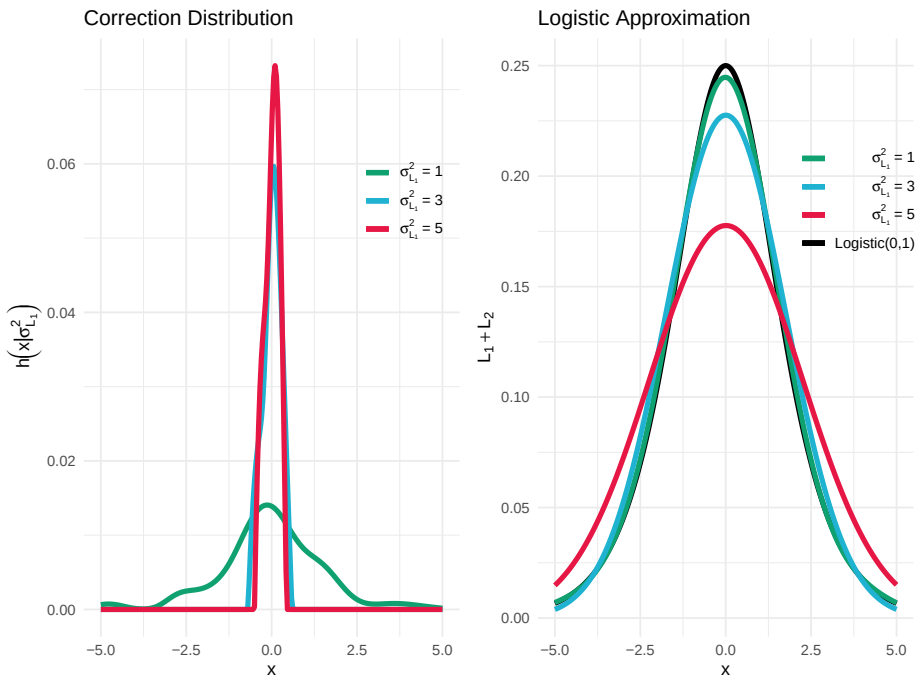
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We set $\sigma^2_{L_1} = (n^2/B)\sqrt{(n-B)/(n-1)}\sigma^2_\Lambda$ then $n\bar{\Lambda}_B$ is equal in distribution to $n\bar{\Lambda}_n + L_1$ so that the acceptance rule in (18) can be rewritten as

$$\begin{aligned}\Delta(\boldsymbol{\theta}_{\text{prop}}, \boldsymbol{\theta}_{\text{cur}}) &= n\bar{\Lambda}_n + \log\left(\frac{\pi(\boldsymbol{\theta}_{\text{prop}})g(\boldsymbol{\theta}_{\text{cur}} \mid \boldsymbol{\theta}_{\text{prop}})}{\pi(\boldsymbol{\theta}_{\text{cur}})g(\boldsymbol{\theta}_{\text{prop}} \mid \boldsymbol{\theta}_{\text{cur}})}\right) + L_1 + L_2 \\ &= n\bar{\Lambda}_B + \log\left(\frac{\pi(\boldsymbol{\theta}_{\text{prop}})g(\boldsymbol{\theta}_{\text{cur}} \mid \boldsymbol{\theta}_{\text{prop}})}{\pi(\boldsymbol{\theta}_{\text{cur}})g(\boldsymbol{\theta}_{\text{prop}} \mid \boldsymbol{\theta}_{\text{cur}})}\right) + L_2. \qquad (22)\end{aligned}$$

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Idea: Choose a cutoff $c \leqslant 3$ and then sample a sufficient minibatch size to ensure $(n^2/B)\sqrt{(n-B)/(n-1)}\sigma^2_\Lambda \leqslant c$.

We then select a minibatch size based on our willingess to inflate the variance.

Algorithm 2



Initialize $\boldsymbol{\beta}_{\text{cur}}$, σ^2_{cur}, $\boldsymbol{\theta}_{\text{cur}}$. Set logistic cutoff $c \leqslant 3$. Estimate correction $h(x \mid c)$.
for i in $1, \dots I$ **do**
    Sample minibatch $\mathcal{B}_\beta$ and draw $\boldsymbol{\beta}$
    Sample minibatch $\mathcal{B}_{\sigma^2}$ and draw σ^2
    Propose $\boldsymbol{\theta}_{\text{prop}} \sim g()$
    Sample $\mathcal{B}_{\boldsymbol{\theta}}$ of size B_{init} from $\{1, \dots, n\}$
    Set $\sigma^2_\Lambda = \mathbb{V}\text{ar}(\{\Lambda_i\}_{i \in \mathcal{B}_{\boldsymbol{\theta}}})$
    while $(n^2/B_{\boldsymbol{\theta}})\sqrt{(n - B_{\boldsymbol{\theta}})/(n-1)}\sigma^2_\Lambda > c$ **do**
        Append $\mathcal{B}_{\boldsymbol{\theta}}$ by sampling an additional B_{inc} from $\{1, \dots, n\} \backslash \mathcal{B}_{\boldsymbol{\theta}}$
        Set $B_\theta \leftarrow B_\theta + B_{\text{inc}}$
        Recalculate $\sigma^2_\Lambda = \text{Var}(\{\Lambda_i\}_{i \in \mathcal{B}_{\boldsymbol{\theta}}})$
    end while
    Draw $L_2 \sim h(\cdot \mid c)$
    Draw $L_1^\star \sim \mathcal{N}(0, c - (n^2/B_{\boldsymbol{\theta}})\sqrt{(n - B_{\boldsymbol{\theta}})/(n-1)}\sigma^2_\Lambda)$
    Calculate

$$\Delta(\boldsymbol{\theta}_{\text{prop}}, \boldsymbol{\theta}) = n\bar{\Lambda}_B + \log\left(\frac{\pi(\boldsymbol{\theta}_{\text{prop}})g(\boldsymbol{\theta} \mid \boldsymbol{\theta}_{\text{prop}})}{\pi(\boldsymbol{\theta})g(\boldsymbol{\theta}_{\text{prop}} \mid \boldsymbol{\theta})}\right) + L_1^\star + L_2$$

    $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta}_{\text{prop}}$ if $\Delta(\boldsymbol{\theta}_{\text{prop}}, \boldsymbol{\theta}) > 0$ using the approximated acceptance rule
    Store $\boldsymbol{\beta}$, σ^2, $\boldsymbol{\theta}$
end for

Simulation Results



Setup:

- 50 simulated data sets.
- $n = 8000$ and $n = 120,000$.
- Locations $\boldsymbol{s}_i$ simulated uniformly on unit square.
- Stationary correlation function with effective range $\sqrt{2}$.

Simulation Results



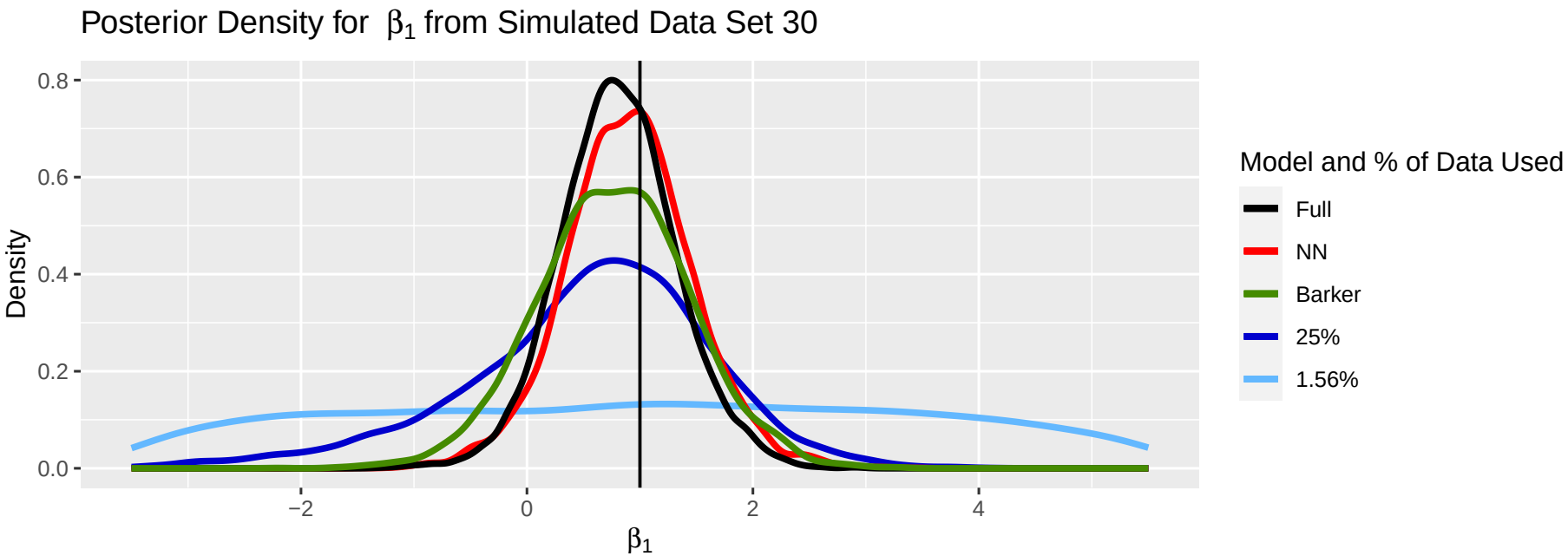
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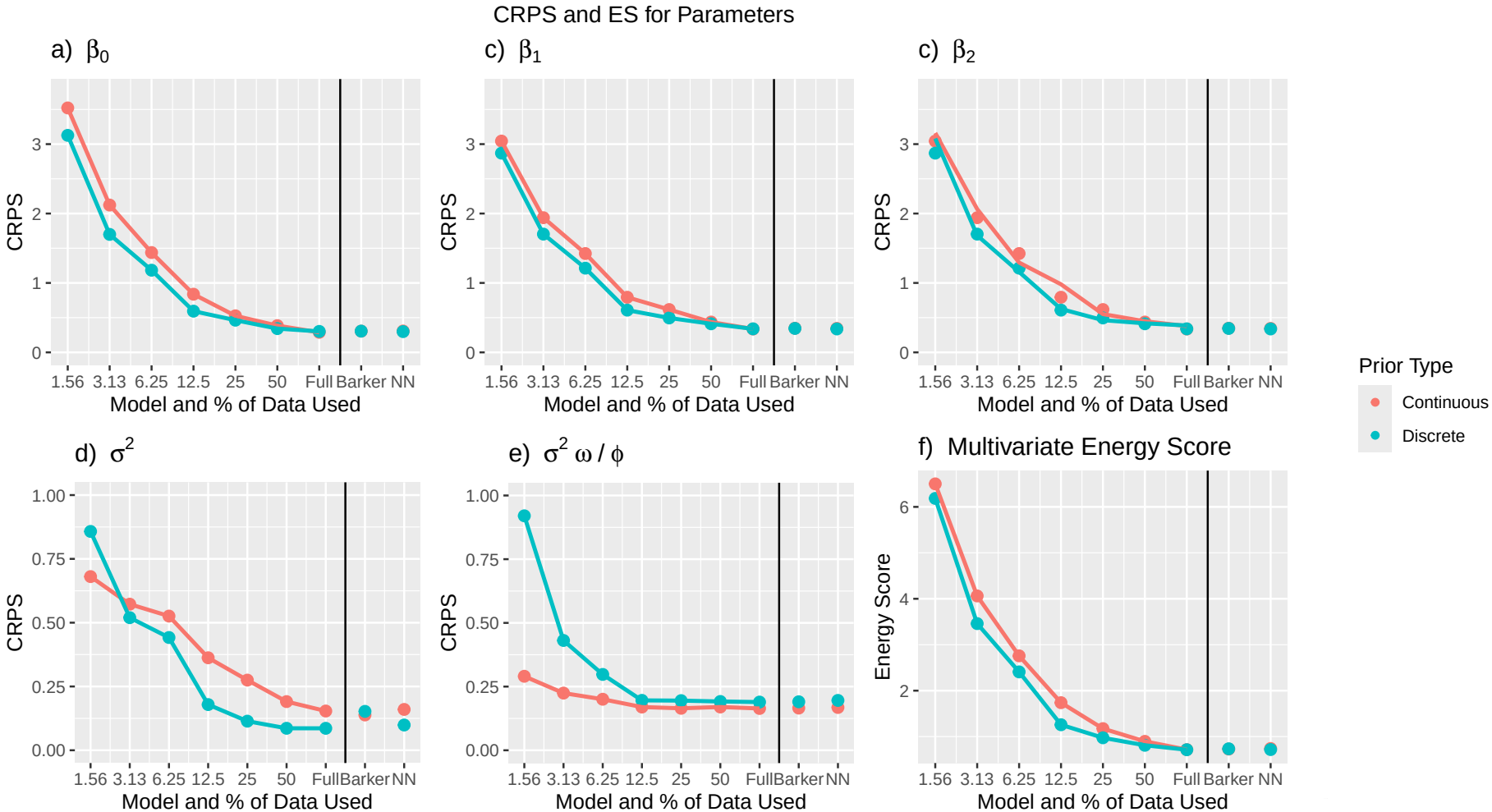
We then compare possible approaches:

- Full GP
- Nearest Neighbor (NN)
- Minibatch MH (FB)
- Minibatch Barker (BG)

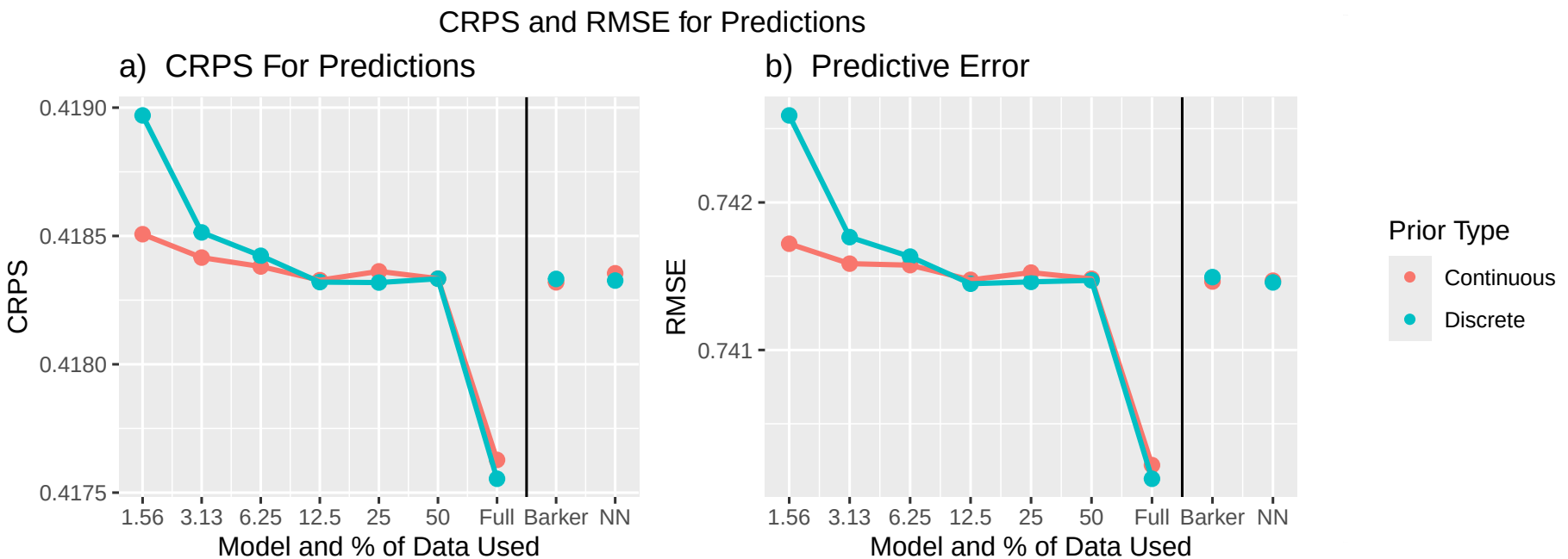
Posterior Comparison and Parameter Inference



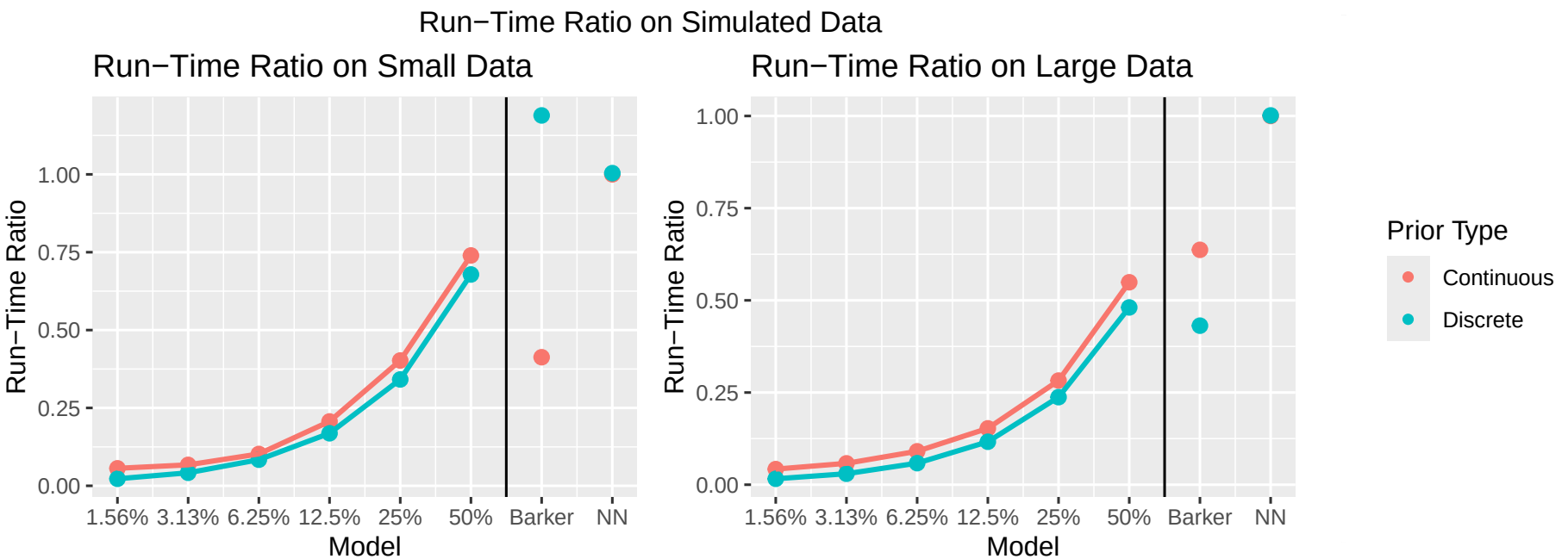
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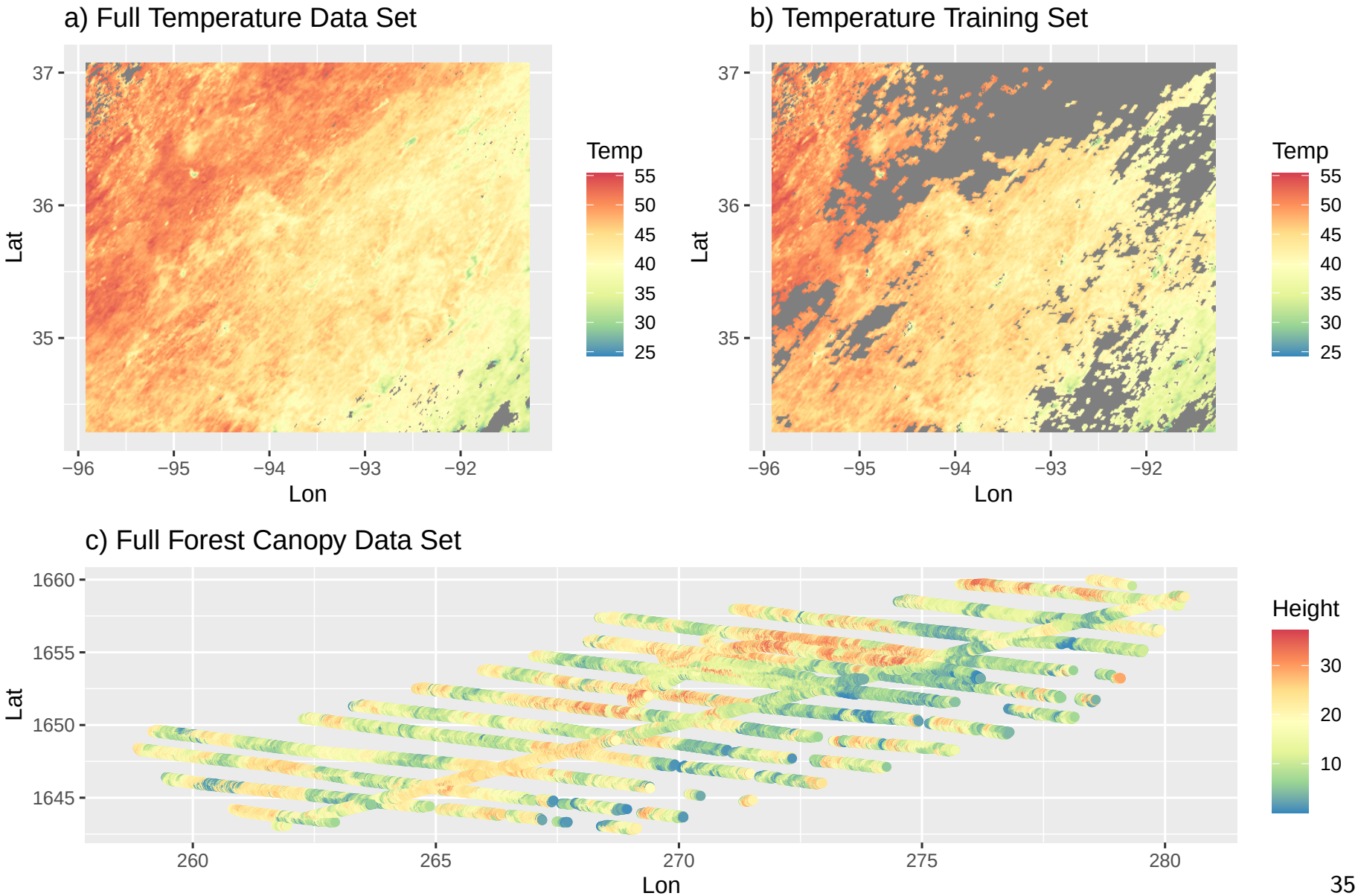
Posterior Predictive Results



Run Time


Run−Time Ratio on Simulated Data
Run−Time Ratio on Small Data
Run−Time Ratio on Large Data
Run−Time Ratio
1.00
0.75
0.50
0.25
0.00
1.56%
3.13%
6.25%
12.5%
25%
50%
Barker
NN
Model
Prior Type
Continuous
Discrete

Application Data and Results



Application Data and Results



	Satellite Data			Forest Data		
Algorithm	Mean	SD	Width	Mean	SD	Width
NN	45.73	0.24	0.87	353.48	4.45	15.41
BG	43.89	2.55	9.28	327.50	17.62	64.35
FB2	45.73	0.64	1.88	353.51	7.20	25.51
FB4	45.71	0.88	2.80	351.85	8.36	29.65
FB8	45.71	1.07	3.73	350.84	10.65	37.51
FB16	45.77	1.41	5.00	351.67	13.39	48.50
FB32	45.67	1.89	7.42	349.57	19.07	69.37
FB64	45.59	3.03	11.37	350.15	27.81	97.65

Posterior Estimates of $\sigma^2\omega/\phi$ using a continuous prior and both application data sets. Note the increase in credible interval width as the minibatch size decreases.

Application Data and Results



Algorithm	MAE	RPMSE	CRPS	INT	WID	CVG
NN	1.38	1.86	0.99	0.23	4.89	0.87
BG	1.36	1.79	0.94	0.19	5.50	0.93
FB2	1.63	2.14	1.22	0.39	4.37	0.74
FB4	1.46	1.95	1.06	0.27	4.70	0.83
FB8	1.47	1.97	1.07	0.28	4.71	0.83
FB16	1.62	2.13	1.21	0.38	4.37	0.75
FB32	1.62	2.12	1.21	0.38	4.39	0.75
FB64	1.46	1.95	1.05	0.25	4.82	0.84

Predictive accuracy for discrete models on the satellite data. These are comparable to the competition numbers shown in Heaton et al. [2019].

Summary



- Developed 2 minibatch approaches for MCMC estimation of Gaussian processes.
  - A fixed batch approach to approximation MH
  - An adaptive approach based on variance inflation
- Minibatching leads to significant computational savings
- Minibatching increases width of posterior credible intervals
- Negligible effect on predictions

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